Complete System Architecture & Executive Guide

Comprehensive Technical Reference · Business Justification · 3-Cycle Automated Reconciliation · Real-World Examples for All Stakeholders

1. Executive Objective — WHY We Are Doing This Activity

Sempra Energy is conducting a large-scale enterprise transformation to SAP S/4HANA across three successive migration waves: **DM1 (Mock 1)**, **DM2 (Mock 2)**, and **DM3 (Dry Run / Cutover)**. Hundreds of millions of business-critical records (Customer Master, General Ledger, Equipment, Service Orders, and Billing) are extracted from legacy ECC systems, transformed, and loaded into SAP S/4HANA.

The Business Problem Solved: Historically, verifying data completeness required large teams of analysts manually merging 6+ massive Excel spreadsheets using complex VLOOKUPs. This manual process took **3 to 4 hours per cycle**, was prone to human copy-paste errors, and lacked real-time visibility into discrepancies.

The Platform Solution: This platform delivers an automated, audit-ready reconciliation engine that ingests, deduplicates, and validates **7,700+ tables in under 10 seconds**. It provides mathematical proof of **Zero Data Loss** and instant governance before cutover sign-off.

■ **<10 Seconds** Full 3-Cycle Sync Time (Was 3+ Hours)	■ **7,700+ Tables** Reconciled End-to-End per Cycle	✔ **100% Automated** Zero Manual Calculation Errors	■ **100% Local** Zero Cloud / Zero License Costs

2. System Architecture & Entire Codebase Breakdown

The platform follows a decoupled, lightweight architecture built for speed, zero licensing cost, and strict on-premise data security:

Component / Layer	File Path & Tech Stack	Functionality & Technical Implementation
Master Cross-Cycle Dashboard	Sempra_CrossCycle_Summary_MASTER.html HTML5, CSS3, JS (ES6+), SheetJS	Single-page executive cockpit. Consolidates DM1, DM2, DM3 with semicircle gauges, DISTINCT aggregation, automated sync, and interactive Src vs Tgt Pivot / Circle modal.
Cycle Dashboards (DM1, DM2, DM3)	Sempra_DM1/2/3_Dashboard.html HTML5, CSS3, JS, Canvas API	Dedicated cycle dashboards for deep-dive functional analysis. Feature table-level search, pagination (500/page), status pills, and direct CSV/Excel exports.
File Watcher Bridge Server	automation/file_watcher_bridge.py Python 3, Flask, Gzip, Threading (Port 5002)	High-performance REST API. Auto-scans local C: drive (.bob/DM1|DM2|DM3) and OneDrive/SharePoint paths. Provides Gzip-compressed byte streaming and ETag 304 caching.
SAP Extraction Engine	automation/sap_bridge.py 01_sap_source_count.py 02_target_count.py	Automates SAP ECC / S4HANA SE16N table count queries via COM GUI Scripting. Implements 3-tier fallback extraction and produces color-coded reconciliation reports.
SharePoint Integration Bridge	automation/sp_bridge.py Python 3, Flask (Port 5004)	Specialized bridge dedicated to serving files directly from SharePoint synced OneDrive directories without local machine path dependencies.
Local Dashboard Launcher	launch_dashboard.py Python 3, HTTP Server (Port 5003)	Launches File Watcher Bridge in background console + spawns local static HTTP server and opens browser automatically at http://localhost:5003/.
Documentation Engine	build_complete_executive_doc.py generate_master_comprehensive_pdf.py	ReportLab PDF & HTML generators that build audit-ready, publication-grade executive documentation with zero external dependencies.

3. WHAT We Can Achieve & What Is Active in the Platform

Active Capability	How It Operates	Business Value & Impact
Dynamic Bridge Resolution	Uses window.location.hostname || 'localhost' for all fetch operations.	Works seamlessly via direct double-click (file://) or server (http://).
Source vs. Target Comparison	Computes Diff = Source - Target with automatic tags: Matching, Source is more, Source is less.	Pre-calculated variance in exported Excel sheets eliminates manual formula errors.
Interactive Circle / Donut View	Multi-segment SVG Donut chart visually categorizing all tables by Source vs Target status.	Gives executives instant visual clarity on overall data reconciliation quality.
Pivot Table & Top Variance Bars	Aggregates record sums and displays horizontal comparison tracks for top 10 variance tables.	Single-click filtering allows engineers to immediately isolate discrepancy tables.
DISTINCT Deduplication Engine	Eliminates duplicate table rows across all 6 EXP and IMP aggregation points.	Guarantees 100% mathematical integrity across all cycle counts.

4. Real-World Practical Examples for ALL Levels of Stakeholders

The platform bridges the gap between executive oversight, functional analysis, technical development, and compliance audit:

■ Level 1: Senior Executives & Steering Committee (10-Second Morning Review)

Workflow: In 10 seconds before a steering committee meeting, the manager clicks **'■ Auto-Sync All Cycles'**. Semicircle health gauges immediately confirm that **DM2 achieved a 99.8% match rate** across 7,715 tables. The manager clicks **'■ Download PDF'** and presents a board-ready report with zero manual spreadsheet preparation.

■ Level 2: Business Data Migration Leads (Granular Discrepancy Resolution)

Workflow: Reviewing Customer Master table /1CADMC/00000239.
• **Source ECC Count:** 12,098,214 records | **Target S/4HANA Count:** 12,098,148 records
• **Dashboard Output:** Tags table automatically as **'▲ Source is more (+66 records)'** with an amber badge.
• **Action:** The lead clicks on the table row, inspects the 66-record gap, and exports a targeted Excel list with pre-calculated diffs for the ETL lead.

■■ Level 3: Technical ETL Developers & SAP Specialists (Automated Extraction)

Workflow: Instead of manually logging into SAP GUI and querying SE16N for 7,700 individual tables, the engineer executes automation/01_sap_source_count.py. The script automates count extraction via COM scripting and streams the data directly to the dashboard cache in seconds.

■■ Level 4: Audit, Risk & Compliance Officers (Verifiable Data Lineage)

Workflow: For SOX and cutover governance, the compliance officer verifies that no records were dropped between Export and Import stages. The officer opens the **'Src vs Tgt Pivot / Circle'** tab, filters by **'✔ Matching'**, and downloads a digitally timestamped Excel audit log proving 100% reconciliation.

5. Complete Workspace Inventory & File Directory

File Name	Category	Lines / Size	Primary Function
`Sempra_CrossCycle_Summary_MASTER.html`	Dashboard	~5,100 lines / 306 KB	Master 3-cycle dashboard with gauges, pivot modal & Excel exports.
`Sempra_DM1_Dashboard.html`	Dashboard	~2,800 lines / 162 KB	Cycle 1 Migration Dashboard with table search & auto-bridge.
`Sempra_DM2_Dashboard.html`	Dashboard	~2,560 lines / 146 KB	Cycle 2 Migration Dashboard with table search & auto-bridge.
`Sempra_DM3_Dashboard.html`	Dashboard	~2,530 lines / 146 KB	Cycle 3 Migration Dashboard with separate EXP/IMP upload.
`Sempra_CrossCycle_Summary_SHAREPOINT.html`	Dashboard	~5,200 lines / 315 KB	SharePoint Edition Master Dashboard with port 5004 mapping.
`automation/file_watcher_bridge.py`	Automation	~450 lines / 16 KB	Flask REST bridge on port 5002 with Gzip & ETag 304 caching.
`automation/sap_bridge.py`	Automation	~590 lines / 29 KB	SAP COM GUI Scripting engine for automated SE16N table counts.
`automation/sp_bridge.py`	Automation	~320 lines / 12 KB	SharePoint folder watcher bridge on port 5004.
`launch_dashboard.py`	Launcher	~85 lines / 3 KB	One-click local launcher for bridge (5002) & web server (5003).
`Sempra_DM_Dashboard_Documentation.pdf`	Deliverable	Multi-page / PDF	Executive & Technical Blueprint PDF for leadership and audit.

6. Step-by-Step User Workflow & Governance

Step 1: Launch	Run python launch_dashboard.py or double-click Sempra_CrossCycle_Summary_MASTER.html.
Step 2: Ingest & Sync	Click '■ **Auto-Sync All Cycles**'. The system ingests and computes reconciliation for DM1, DM2, and DM3 in <10s.
Step 3: Analyze Gaps	Click on any count number (e.g. **7,710**) to inspect individual tables or switch to '■ **Src vs Tgt Pivot / Circle**'.
Step 4: Export Reports	Click ■ **Excel** to export audit-ready spreadsheets with pre-calculated Source/Target diff and status columns.